

From Microsociology to VR Design: A Framework for integrating Non-Verbal Behavior in Adaptive Systems

Iryna Ignatieva*

Vision and Imaging Technologies; The Interactive and
Cognitive Systems Group
Fraunhofer Heinrich-Hertz-Institute
Berlin, Germany
iryna.kuptsova@hhi.fraunhofer.de

Karam Tomotaki-Dawoud*

Vision and Imaging Technologies; The Interactive and
Cognitive Systems Group
Fraunhofer Heinrich-Hertz-Institute
Berlin, Berlin, Germany
karam.tomotaki-dawoud@hhi.fraunhofer.de

Birgit Nierula*

Vision and Imaging Technologies; The Interactive and
Cognitive Systems Group
Fraunhofer Heinrich-Hertz-Institute
Berlin, Germany
birgit.nierula@hhi.fraunhofer.de

Sebastian Bosse*

Vision and Imaging Technologies; The Interactive and
Cognitive Systems Group
Fraunhofer Heinrich-Hertz-Institute
Berlin, Germany
sebastian.bosse@hhi.fraunhofer.de

Abstract

Adaptive virtual reality (VR) training systems face a fundamental challenge: nonverbal cues are context-dependent and resist straightforward interpretation. We present a framework that translates microsociological theory into practical system design through two phases: dataset development (defining meaningful gestures/postures and related scenarios (embedding cues in realistic situations), and perception studies (measuring interpretation variance). As a test-bed, we apply this framework to police-citizen conflict resolution training - a domain lacking design-oriented empirical data. We identified 19 gestures/postures across 10 scenarios and collected perception data from 600 residents in Germany. This work demonstrates a replicable methodology for developing realistic adaptive systems. The framework addresses key limitations in current approaches with applications extending beyond police training to any domain requiring sophisticated behavioral interpretation of conflict interactions.

CCS Concepts

• **Human-centered computing** → **User centered design; Contextual design; Scenario-based design; Interaction design theory, concepts and paradigms; Virtual reality**; • **Applied computing** → **Sociology; Interactive learning environments**.

Keywords

virtual reality, adaptive systems, microsociology, conflict resolution, VR training systems, human-computer interaction

*These authors contributed equally.



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ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798627>

ACM Reference Format:

Iryna Ignatieva, Birgit Nierula, Karam Tomotaki-Dawoud, and Sebastian Bosse. 2026. From Microsociology to VR Design: A Framework for integrating Non-Verbal Behavior in Adaptive Systems. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798627>

1 Introduction

Modern adaptive VR training systems promise genuine interaction through real-time responsiveness, yet struggle to interpret nonverbal behavior realistically. The reason is straightforward: nonverbal behavior is inherently challenging to interpret. While some gestures can be interpreted relatively consistently within and even across cultures [14], it is a misconception to assume that all nonverbal cues carry universal or fixed meanings [17]. Instead interpretation is shaped by social context, situational dynamics, participants' characteristics, and communicative goals - complexities rarely addressed in adaptive VR training systems. To address this gap, we propose a methodological framework for translating microsociological insights on nonverbal behavior into adaptive VR training system design. We illustrate it through a VR police training use case on conflict resolution, where the lack of design-oriented empirical data on gestures and postures - particularly their escalation or de-escalation potential - posed a key challenge. Serving as a testbed, this case offers a replicable model for integrating microsociological insights into adaptive VR systems design.

We begin by reviewing current approaches and their limitations in interpreting nonverbal behavior for VR training systems. Next, we establish microsociological foundations for understanding conflict dynamics. We then present our framework and demonstrate it through a police conflict resolution training case study.

2 Related work

Interpreting nonverbal behavior in VR systems requires translating the complexity of real-life social interactions into computational algorithms. Current approaches fall into two main categories, each with distinct limitations.

Theory-driven approaches use social science frameworks to interpret nonverbal cues. For instance, the established link between open body postures and engagement could be modeled using Bayesian networks. [1]. A key limitation is that social science findings are applied directly to computational models, despite the original research not being designed for this purpose. While general associations are useful, they rarely account for individual, situational, or cultural variance.

Data-driven approaches rely on real-life interaction videos [2], but obtaining valid training datasets for conflict interactions poses significant challenges [20]. Social signal processing often relies on publicly available datasets, for example, on videos of political debates [21] [10]. However, domains like police–citizen interactions face a dual problem. Less biased datasets from body-cameras are restricted by access policies, while publicly available videos are biased toward escalated conflicts. Accessible datasets may not capture the full range of situations—from de-escalation to high-intensity conflict—needed for realistic VR scenarios and nuanced gesture interpretation.

State of the art adaptive systems rely on metrics focusing only on the VR user, such as performance metrics or physiological markers [4] [9], rather than focusing on the trainee-agent dynamics of the interaction. For example, in high-stress police training, trainers can modify scenario complexity according to trainees’ measured stress levels [22]. Due to the challenges associated with interpretation of nonverbal behavior, such cues remain largely unused in system design. As a result VR adaptive systems rely on more easily measurable indicators, such as heart rate, spatial proximity, and movement patterns [15].

Drawing on microsociological foundation, we examine strategies that go beyond capturing user metrics and allow to investigate the dynamics of user-agent interaction. Focusing on user-avatar interaction allows training narratives to evolve based on interactional dynamics, creating more realistic and immersive experiences.

3 Microsociology of conflict as a theoretical framework for conflict resolution VR training

In face-to-face interactions, meaning emerges through participants’ interpretive processes, with nonverbal cues shaped by context, situation, individual traits, and communicative goals [3] [17]. Microsociologists studying conflict interactions identify situational dynamics as the key predictor of escalation, emphasizing their central role in determining when and how violence emerges [7] [13]. In conflict, most participants are constrained by “confrontational tension and fear,” with violence occurring only when one party gains situational advantage and emotional dominance [7]. Research on police–protester confrontations supports this perspective, identifying “situational breakdown” - marked by escalating tension and overwhelming fear - as a key precursor to violence [16].

Violence is one extreme of the escalation spectrum, but tension can also rise, decline, or resolve non-confrontationally. Microsociology examines interaction through turn-taking, where each response shapes the next [18] [13], and turning points, abrupt shifts that redirect the interaction [6]. In police–citizen encounters, turning points may involve sudden changes in non-verbal behavior that require immediate responses, including the use of force [12]. This perspective

highlights not only when violence occurs but also how situational dynamics shift and what triggers these changes. Turning points can also lead to de-escalation, occurring when participants connect despite their differences and ongoing conflict [11] [8] [5].

The microsociological perspective provides key insights into how interactions unfold in real time. From this foundation, we highlight three principles for system design: (1) nonverbal communication is meaningful only within specific scenarios, (2) situational dynamics emerge as participants continuously interpret and respond to one another, and (3) the trajectory of confrontational tension - escalating or de-escalating - is shaped by turning points marked by shifts in nonverbal behavior. These insights inform system design by linking nonverbal cues to key micro-moments and treating interactions as dynamic, co-constructed processes.

In the next paragraph, we propose a framework for computationally extracting and interpreting nonverbal cues in social interactions for adaptive VR training systems

4 A Framework for Extracting Computable Social Interaction Data

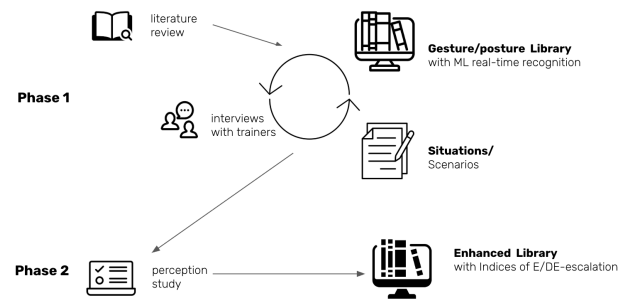


Figure 1: A Framework for Extracting Computable Social Interaction Data

The proposed framework collects data for adaptive interaction logic through two phases: First phase, *Dataset development*, defines gestures and postures meaningful for the system’s functionality, translating implicit human behavior into computational categories and identifies realistic scenarios in which these gestures occur. Second phase, *Perception study*, measures how interpretation of these gestures varies within scenarios, translating findings into a form that can be integrated into the system (Fig. 1).

This multi-step process centers on data triangulation: moving from theoretical frameworks to expert knowledge to empirical perception data. Unlike conventional approaches assuming fixed meanings, our framework grounds interpretation in actual perceptual responses. This shifts VR design toward modeling realistic reactions at behavioral turning points rather than simply calibrating challenge levels based on performance.

4.1 Case Study: Adaptive VR for Police Conflict Resolution Training

We demonstrate framework viability through a case study developing an adaptive VR training system for police conflict resolution.

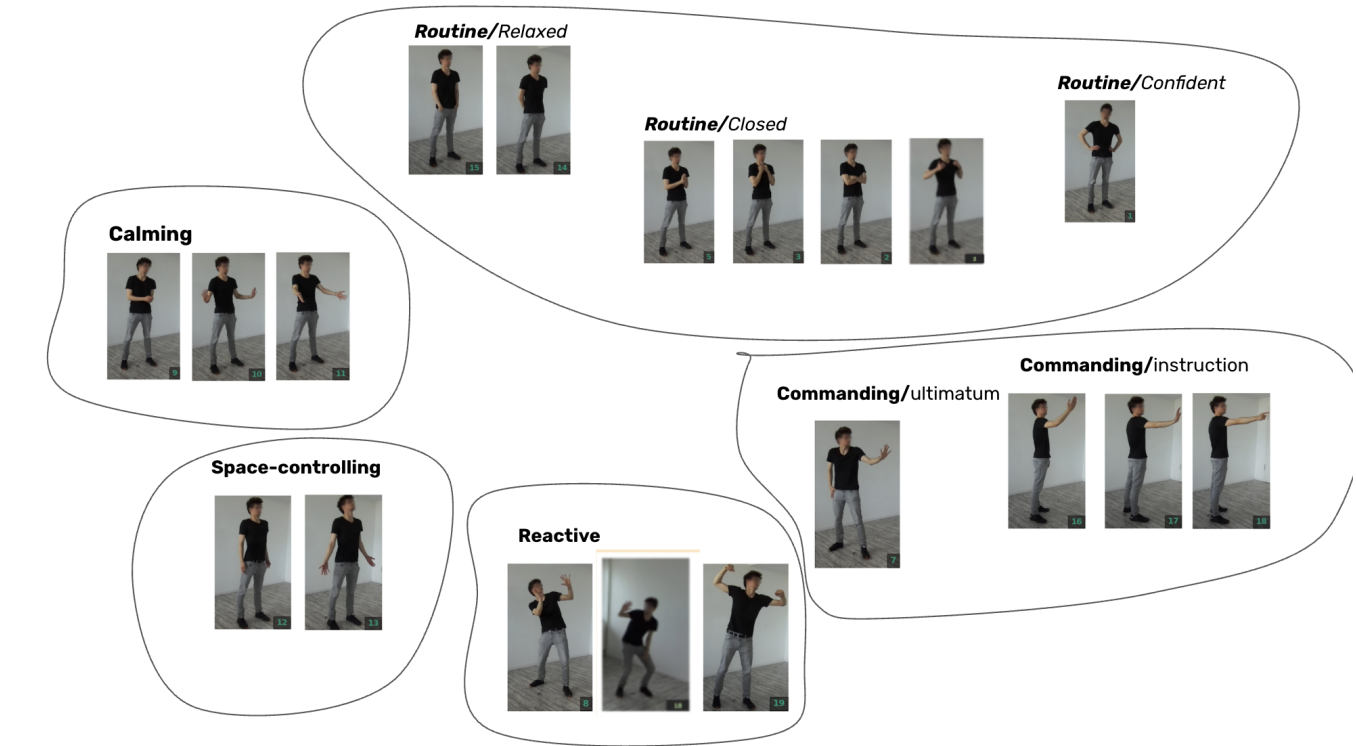


Figure 2: Gesture/posture library categorized by communicative function

This domain presents unique challenges: no existing datasets and privacy restrictions prevent access to real interaction videos. The case study serves as a testbed for the methodological approach while addressing a practical training need.

4.1.1 Dataset development. We identified specific nonverbal communicative markers influencing situational dynamics. Based on literature review and 2 expert interviews with police trainers, we identified 19 gestures and postures categorized by communicative function (routine/alert, calming, commanding, space-controlling, reactive) (Fig. 2) and developed a gesture/posture library with machine learning pipeline for real-time detection and classification [19].

Then, based on five elicitation interviews with trainers, we defined 10 realistic scenarios. Trainers assessed gestures and postures from the gesture/posture library - performed by male and female actors playing officers - on a 10-point escalation/de-escalation scale, suggested possible real-life situations, and noted how interpretation might vary by context. Scenarios were selected based on realism, commonality in public police-citizen interactions, professional officer conduct and potential for escalation. Scenarios were grouped by escalation scores: Type A (low tension) and Type B (high tension). Some gestures applied to both types depending on context; others fit only one. Consistent with microsociological studies, Type B (high escalation) was defined by turning points where citizens break expected interaction patterns, requiring police to react. For example, a driver pulled over for running a red light denies the violation and argues instead of following the standard protocol.

4.1.2 Perception study. We conducted an online study to explore how residents in Germany ($N = 600$) perceive police officers' gestures and body postures in the scenarios defined by police trainers.

Participants. Participants were recruited through Billendi's online panel using non-probability quota sampling stratified by age and gender to enable demographic-specific analysis. Migration background treated as controlled variable, included proportionally to match German population demographics. Informed consent was obtained prior to the experiment, and the protocol was approved by the Ethics Committee.

Stimuli. Short video clips (2–5 sec.) featured male and female actors in authentic police uniforms; faces were digitally blurred against a uniform gray background (Fig. 3 left). Each gesture was embedded in a situational narrative — one or two scenarios (Types A and B) depending on gesture type. Routine, commanding, and calming gestures shared a subgroup-level scenario, while remaining gestures had gesture-specific scenarios. Clips featuring male and female officers were balanced across all 19 gestures, resulting in 60 clips per participant (Fig. 3 left).

Procedure. First, participants received operational definitions: escalation was defined as nonverbal behavior that increases tension, aggression, or conflict potential, while de-escalation referred to body language that facilitates situational calming and tension reduction. Participants read scenario descriptions, then viewed short video clips depicting relevant gestures. For each clip, they assessed whether an officer's nonverbal behavior would escalate or de-escalate the situation using a continuous scale from -1000

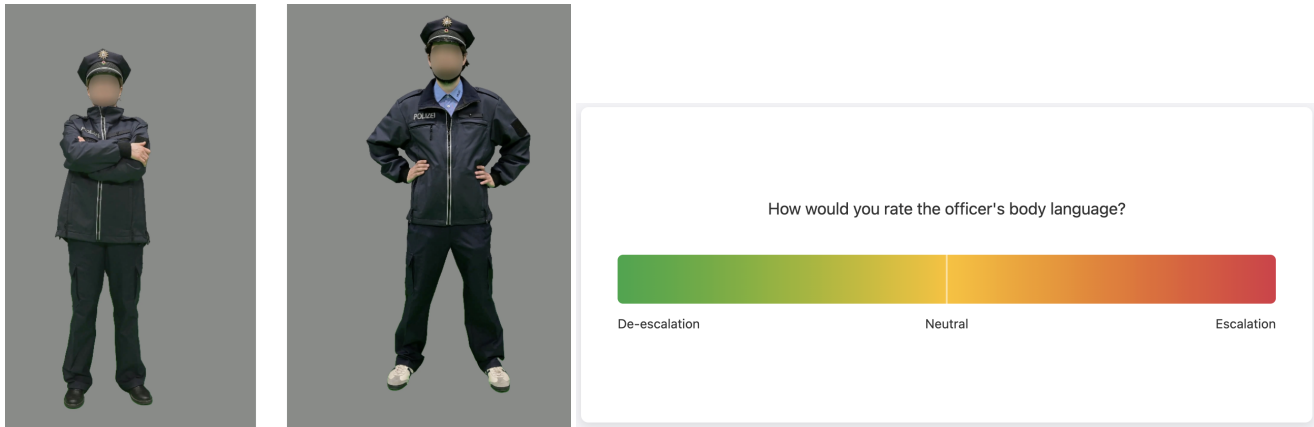


Figure 3: Screenshot of the online study: Left, displaying male and female police officers (stimuli). Right, interface showing the escalation/de-escalation rating slider.

(maximum de-escalation) to +1000 (maximum escalation), neutral midpoint at 0 (Fig. 3 right). Numerical values were concealed to eliminate strategic responding. Participants were instructed to respond quickly after each clip - time pressure designed to simulate realistic split-second decision-making in actual encounters, replicating conditions where citizens must process multiple information streams while maintaining situational awareness.

Results. The study will generate escalation/de-escalation indices for each gesture within each scenario type, differentiated by officer gender and citizen demographics (age, gender). These indices will serve as computational parameters for adaptive VR system design. Data collection is complete, and analysis is currently in progress.

Preliminary analysis confirms the framework captures intended perceptual patterns. For example, calming gestures—'Head tilting', 'Empty hands', and 'Opening hands'—within a single scenario were consistently rated as de-escalating (mean slider values below zero), validating our design approach.

This same scenario also reveals substantial interpretation variance across demographics. Older participants (35+) rated these gestures as even more de-escalating than younger participants (18–34), with differences statistically significant in post-hoc Tukey HSD tests ($p < 0.05$).

Scenario types A and B also showed perceptual differences. Slider ratings for the commanding gestures "Beckoning someone," "Stopping," and "Pointing" differed significantly between scenarios (Mann-Whitney $U = 6,175,266.5$, $p = .005$, rank-biserial $r = 0.038$).

These findings demonstrate the framework's ability to identify perceptual differences critical for adaptive system design.

5 Discussion

Proposed framework functions in domains where video datasets are unavailable or difficult to access. This makes the approach viable for privacy-sensitive domains. The resulting computational indices - calibrated to reflect different de-/escalation potentials across scenario types - are designed provide concrete parameters for VR system adaptation, moving beyond universal gesture meanings toward situation-framed interpretation.

Quota-based sampling, mirroring population demographics, enables analysis of perceptual group effects. This supports designing less biased systems sensitive to how different groups perceive non-verbal behavior. Upon implementation, it could enable interactional dynamics based on both virtual character and trainee profiles.

Another advantage of the proposed framework is its practical feasibility: it is cost-effective and easy to implement, making it replicable across different training systems. While demonstrated in a police training context, the framework is not domain-specific. Its transferability rests on the escalation–de-escalation logic and the turning point as a core analytical unit of conflict interaction — concepts that apply wherever practitioners must read and respond to behavioral cues in real time across high-stakes interpersonal encounters.

With additional resources, greater precision can be achieved by making microsociological studies of domain-specific scenarios a first step in development. The perception study could be extended through more representative sampling or expanded beyond age and gender categories, depending on the training objectives.

Since nonverbal cues are interpreted differently across cultural contexts, we developed the framework within a single cultural setting. The proposed approach treats cultural awareness as a methodological advantage central to realistic interaction modeling. However, different cultural perspectives can be incorporated by replicating the framework in multiple contexts, allowing the same system architecture to support multiple cultural modes.

6 Limitations and Future Work

This work presents a methodological framework with initial implementation; the case study remains in the analysis stage. Full validation requires completing the data analysis, implementing the adaptive VR system, and conducting user studies to demonstrate improved training outcomes.

The framework currently addresses nonverbal behavior only; integrating additional modalities into a multimodal system represents a necessary next step. While we describe the framework

for VR adaptation, actual system implementation and validation of training effectiveness constitute future work.

7 Conclusion

We present a theoretical framework that bridges microsociology and adaptive VR system design, addressing a fundamental challenge: how to interpret context-dependent nonverbal behavior in interactive systems. Proposed framework grounds behavioral interpretation in empirical perception data collected within specific situational contexts. By measuring how interpretation varies across scenarios and demographic groups, the framework generate computational indices that capture the nuanced, context-sensitive nature of human communication. This enables virtual agents to respond realistically based on both the agent's profile and the trainee's characteristics, reflecting how these factors shape real-world interactions.

This work moves beyond simplistic behavioral models toward embracing interaction complexity. By providing gesture/posture datasets with context-specific escalation/de-escalation indices, we offer concrete resources enabling researchers and practitioners to build more socially intelligent VR training environments.

Once validated, this framework enables: (a) adaptation logic based on realistic interactional responses rather than performance metrics, (b) dynamic narrative adjustment through virtual agents that escalate or de-escalate based on trainee gestures, and (c) profile-based responses that vary according to both virtual character and trainee demographics.

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